The stylized model, exactly as it is coded

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Demography and timing

Overlapping generations, two periods of thirty years, ages 25–55 and 55–85. Cohort mass one. A household has one child *in period 1 only*; the child leaves at the end of it and enters as a young adult, so the equivalence scale is $\xi = \sqrt{2}$ in period 1 and $\xi = 1$ in period 2. Every annual parameter is compounded to thirty years: $\beta = 0.995^{30} = 0.86038$, and $R = 1$, so a unit saved is a unit carried.

Period 1

State. Permanent income y (lognormal, 120 nodes). An employment draw (unemployed with probability u_0) and, if income-eligible, a voucher draw (probability π_V) give realised income and a rent multiplier (inc, r). Income is floored, $\text{inc} \leftarrow \max(\text{inc}, \underline{y})$. The means test is $\text{inc} \le y_V$; a voucher cuts the tenant's rent to κ_r of the market rent.

Choices. Drug purchase $v \ge 0$, housing $h \in \{h_{lg}, h_{sm}, h_{hl}\}$ (large unit, small unit, street — the street is free and supplies h_{hl}), and saving $a' \ge 0$, subject to

$$c = \text{inc} - p_v v - \text{rent}(h, r) - a' \;>\; 0.$$

Flow utility. With $\varphi_h = \gamma(1-\theta)$, $\varphi_c = (1-\gamma)(1-\theta)$, $\varphi_q = \theta$ and $u(x) = x^{1-\sigma}/(1-\sigma)$, $\sigma = 2$:

$$X_P = \left(\tfrac{c}{\xi}\right)^{\varphi_c}\left(\tfrac{h}{\xi}\right)^{\varphi_h}(\bar{b} + v)^{\varphi_q}, \qquad X_K = \left(\tfrac{c}{\xi}\right)^{\varphi_c}\left(\tfrac{h}{\xi}\right)^{\varphi_h}\bar{b}^{\,\varphi_q},$$

$$\text{flow} \;=\; u(X_P) \;+\; \omega_c(\xi - 1)\, u(X_K) \;+\; \aleph.$$

c and h are divided by ξ; v is not, because the child consumes none of it. The child's bundle carries the Stone–Geary shifter $\bar{b}$ alone, so the drug reaches the child only by taking consumption out of the shared c. $\aleph$ is added once per household, not once per member.

The transition. The disorder is realised on the way to period 2 with

$$\pi(v) = \pi_{\text{base}} \cdot \mathbf{1}\{v > 0\}$$

— flat in quantity, so any positive purchase carries the full onset risk. Survival is $s_0 = \phi_0$ clean and $s_1 = \phi_0(1-\kappa)$ with the disorder.

The problem. Choose (v, h, a') to maximise

$$\boxed{\mathcal{V} = u(X_P) + \omega_c(\xi - 1)u(X_K) + \aleph \;+\; \beta\Big[(1 - \pi(v))\, s_0\, W_2(a', y, 0) + \pi(v)\, s_1\, W_2(a', y, 1)\Big] \;-\; \beta\, \omega_c\, \Delta(y)\, \pi(v)}$$

Survival sits *inside* the expectation, not outside: $\phi\, \mathbb{E}W_2$ makes the disorder desirable wherever $W_2 < 0$. The $v = 0$ branch is solved separately, on the clean continuation and with the voucher unrevoked, and the household uses if the best $v > 0$ beats it.

Period 2 — terminal

Employment is redrawn (u_0 clean, u_1 with the disorder) and the voucher is redrawn subject to the policy rule. Assets are run down and nothing is bequeathed, so with $\text{cash} = Ra + \text{inc}$ and $\xi = 1$:

$$W_2(a, y, d) = \max_h \; u\big(c^{\varphi_c} h^{\varphi_h} \bar{b}^{\,\varphi_q}\big) + \aleph, \qquad c = \text{cash} - \text{rent}(h, r).$$

No drug choice, no child term, no recovery: mortality taxes a continuation, so the disorder can only cost anything if realised at a transition rather than inside the terminal period.

Policy. Housing First pays the voucher to every eligible household; Treatment First withdraws it from any household failing the sobriety condition *in the period the voucher is paid* — current use or a current disorder. That single difference is the whole experiment.

The child

The child inherits income by persistence, with the innovation that keeps the cross-section stationary:

$$\log y_c = \rho \log y + (1-\rho)\log \bar{y} - \delta_d \mathbf{1}\{d' = 1\} + \varepsilon, \qquad \varepsilon \sim N\big(0, (1-\rho^2)\sigma_y^2\big), \quad \rho = 0.5.$$

There is *no* homelessness penalty: persistence alone delivers that odds ratio, so δ_d is the only inherited penalty and the mixture has two types, not four. The child is run through the model as an adult at its own income, which is what gives it a disorder probability at all. $\Delta(y)$ is what the parent's disorder costs the child — the difference, not the level:

$$\Delta(y) = \mathbb{E}\big[V_1(y_c) \,\big|\, \delta_d = 0\big] - \mathbb{E}\big[V_1(y_c) \,\big|\, \delta_d\big],$$

both over the same kernel. V_1 is on both sides, so this is a fixed point, solved by iteration; it is well behaved because the $\aleph$ levels cancel in the difference and the term enters multiplied by $\beta\omega_c\pi(v)$.

Lifetime utility

$V_1(y)$ is the household's *own* value: the maximised $\mathcal{V}$ with the internalised child term $\beta\omega_c\Delta(y)\pi(v)$ **added back**, so a planner counts the child once rather than once in full and again at ω_c. The childhood a cohort actually lives, at its parent's realised allocation, is

$$W_{\text{child}} = (\xi - 1)\big[u(X_K) + \aleph\big],$$

with the $\aleph$ term a switch: a childhood is a period of life, but $\aleph$ is added once per household rather than per member. Lifetime utility is

$$W = W_{\text{child}} + \beta\, V_1,$$

childhood lived under the policy in force one period earlier and adulthood under the policy in force now. **The code currently forms $W_{\text{child}} + V_1$ with no β**, which was invisible while $\beta = 1$ and is not now; flagged, not changed.

Consumption equivalents are CRRA, not log. Scaling c by $(1+\lambda)$ scales both bundles by $(1+\lambda)^{\varphi_c}$, so every u term scales by $(1+\lambda)^{\varphi_c(1-\sigma)}$ while the $\aleph$ terms do not move:

$$\lambda = \left(1 + \frac{dV}{U}\right)^{\frac{1}{\varphi_c(1-\sigma)}} - 1, \qquad U = V - A, \qquad A = \aleph\Big(1 + \beta\big[(1-\pi)s_0 + \pi s_1\big]\Big).$$

It depends on the *level* through U, which the log formula never did.

Parameters

set		calibrated	on
$\sigma = 2$ risk aversion	$\rho = 0.50$ persistence	γ	housing / non-drug spending
$\beta = 0.995^{30} = 0.86038$	$R = 1$	$\aleph$	value of a statistical life
$\phi_0 = 0.91$ survival	u_0 unemployment	h_{hl}	homelessness rate
κ from the SMR	$\underline{y}$ income floor	$\bar{b}$	participation
h_{lg}, h_{sm} unit sizes	κ_r, y_V, π_V voucher	θ	drug spending share
$n_a = 60$, $n_v = 140$ grids	$v_{lo} = 10^{-5}$	π_{base}	disorder rate
		ω_c	child use gap
		δ_d	child SUD odds ratio

Identity: sud = participation $\times\, \pi_{\text{base}}$. ω_c acts in two places, the flow and the continuation; the continuation half was identically zero until $\Delta(y)$ was built.